

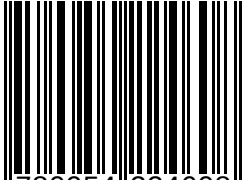
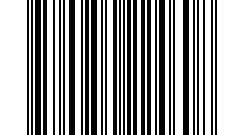
Barcodes

NB **Quiet zones** - The barcode object includes space to the right/left or top/bottom only when the specification states a 'quiet zone' or 'light margin'. All the examples below also have CSS property set on the barcode object i.e. padding: 1.5mm;

EAN-13 Barcodes (EAN-2 and EAN-5)

NB EAN-13, UPC-A, UPC-E, and EAN-8 may all include an additional bar code(EAN-2 and EAN-5) to the right of the main bar code (see below).

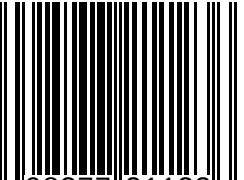
A nominal height and width for these barcodes is defined by the specification. 'size' will scale both the height and width. Values between 0.8 and 2 are allowed (i.e. 80% to 200% of the nominal size). 'height' can also be varied as a factor of 1; this is applied after the scaling factor used for 'size'.

CODE	DESCRIPTION	BARCODE
EAN13	Standard EAN-13 barcode. Accepts 12 or 13 characters (creating checksum digit if required). [0-9] numeric only.	9 7 8 - 0 - 9 5 4 2 2 4 6 - 0 - 8  9 780954 224608 >
ISBN	Standard EAN-13 barcode with 'ISBN' number shown above [shown at height="0.66"]	ISBN 978-0-9542246-0-8  9 780954 224608 >
ISSN	Standard EAN-13 barcode with 'ISSN' number shown above [shown at size="0.8"]	ISSN 978-0-9542246-0-8  9 780954 224608 >

EAN-8, UPC-A and UPC-E Barcodes

UPC-A, UPC-E, EAN-13, and EAN-8 may all include an additional bar code(EAN-2 and EAN-5) to the right of the main bar code (see below).

A nominal height and width for these barcodes is defined by the specification. 'size' will scale both the height and width. Values between 0.8 and 2 are allowed (i.e. 80% to 200% of the nominal size). 'height' can also be varied as a factor of 1; this is applied after the scaling factor used for 'size'.

CODE	DESCRIPTION	BARCODE
UPCA	UPC-A barcode. This is a subset of the EAN-13. (098277211236) Accepts 11 or 12 characters (creating checksum digit if required). [0-9] numeric only	 0 98277 21123 6

CODE	DESCRIPTION	BARCODE
UPCE	UPC-E barcode. Requires the UPC-A code to be entered as above (e.g. 042100005264 to give 425261). NB mPDF will die with an error message if the code is not valid, as only some UPC-A codes can be converted into valid UPC-E codes. UPC-E doesn't have a check digit encoded explicitly, rather the check digit is encoded in the parity of the other six characters. The check digit that is encoded is the check digit from the original UPC-A barcode.	 0 425261 4
EAN8	EAN-8. Accepts 7 or 8 characters (creating checksum digit if required). [0-9] numeric only	 < 2468 1230 >

EAN-2 and EAN-5 supplements, and combined forms

UPC-A, UPC-E, EAN-13, and EAN-8 may all include an additional bar code (EAN-2 and EAN-5) to the right of the main bar code.

CODE	DESCRIPTION	BARCODE
EAN2	EAN-2 supplement barcode. mPDF does not generate EAN-5 barcode on its own; see supplements below. Used to denote an issue of a periodical. EAN-2 supplement accepts 2 digits [0-9] only, EAN-5 five.	
EAN5	EAN-5 supplement barcode. mPDF does not generate EAN-5 barcode on its own; see supplements below. Usually used in conjunction with EAN-13 for the price of books. 90000 is the code for no price.	
EAN13P 2	Standard EAN-13 barcode with 2-digit UPC supplement (07)	 9 780954 224608 07 >
ISBNP2	Standard EAN-13 barcode with 'ISBN' number shown above, and 2-digit EAN-2 supplement	ISBN 978-0-9542246-0-8  9 780954 224608 07 >
ISSNP2	Standard EAN-13 barcode with 'ISSN' number shown above, and 2-digit EAN-2 supplement	ISSN 978-0-9542246-0-8  9 780954 224608 07 >
UPCAP2	UPC-A barcode with 2-digit EAN-2 supplement. This is a subset of the EAN-13.	 0 06338 95260 8 24 >

CODE	DESCRIPTION	BARCODE
UPCEP2	UPC-E barcode with 2-digit EAN-2 supplement.	
EAN8P2	EAN-8 barcode with 2-digit EAN-2 supplement	
EAN13P5	Standard EAN-13 barcode with 5-digit UPC supplement (90000)	
ISBNP5	Standard EAN-13 barcode with 'ISBN' number shown above, and 5-digit EAN-5 supplement	ISBN 978-0-9542246-0-8 
ISSNP5	Standard EAN-13 barcode with 'ISSN' number shown above, and 5-digit EAN-5 supplement	ISSN 978-0-9542246-0-8 
UPCAP5	UPC-A barcode with 5-digit EAN-5 supplement. This is a subset of the EAN-13	
UPCEP5	UPC-E barcode with 5-digit EAN-5 supplement. (042100005264 90000)	
EAN8P5	EAN-8 barcode with 5-digit EAN-5 supplement (55123457 90000)	

Postcode Barcodes

These all have sizes fixed by their specification. Although they can be altered using 'size' it is not recommended. 'height' is ignored.

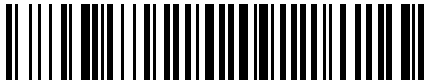
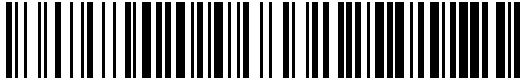
CODE	DESCRIPTION	BARCODE
IMB	Intelligent Mail Barcode - also known as: USPS OneCode 4-State Customer Barcode, OneCode 4CB, USPS 4CB, 4-CB, 4-State Customer Barcode, USPS OneCode Solution Barcode. (01234567094987654321-01234567891) Accepts: Up to 31 digits (required 20-digit Tracking Code, and up to 11-digit Routing Code; this may be 0, 5, 9, or 11 digits). If the Routing code is included, it should be separated by a hyphen - like this example.	
RM4SC C	Royal Mail 4-state Customer barcode (SN34RD1A). Accepts: max. 9 characters. Valid characters: [A-Z,0-9] Checksum digit: automatic.	
KIX	Dutch KIX version of Royal Mail 4-state Customer barcode (SN34RD1A). Valid characters: [A-Z,0-9]. Checksum digit: automatic.	
POSTN ET	POSTNET barcode. Accepts 5, 9 or 11 digits. Valid characters: [0-9]. Checksum digit: automatic.	
PLANET	PLANET barcode. Accepts 11 or 13 digits. Valid characters: [0-9]. Checksum digit: automatic.	

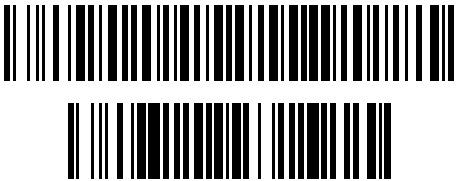
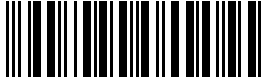
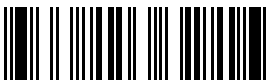
Variable width Barcodes

These barcodes are all of variable length depending on the code entered. There is no recommended maximum size for any of these specs, but all recommend a minimum X-dimension (width of narrowest bar) as 7.5mil (=0.19mm). The default used here is twice the minimum i.e. X-dim = 0.38mm.

The specifications give a minimum height of 15% of the barcode length (which can be variable). The bar height in mPDF is set to a default value of 10mm.

'size' will scale the barcode in both dimensions. mPDF will accept any number, but bear in mind that size="0.5" will set the bar width to the minimum. The 'height' attribute further allows scaling - this factor is applied to already scaled barcode. Thus size="2" height="0.5" will give a barcode twice the default width (X-dim=0.76mm) and at the default height set in mPDF i.e. 10mm.

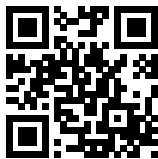
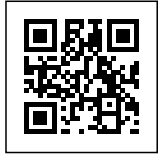
CODE	DESCRIPTION	BARCODE
C128A	CODE 128 A. Valid characters: [A-Z uppercase and control chars ASCII 0-31]. Checksum digit: automatic.	
C128B	CODE 128 B. Valid characters: [Upper / Lower Case + All ASCII Printable Characters]. Checksum digit: automatic.	
C128C	CODE 128 C. Valid characters: [0-9]. Must be an even number of digits. Checksum digit: automatic.	

CODE	DESCRIPTION	BARCODE
C128RA W [A/B/C]	CODE 128 A/B/C symbols. You can generate any barcode (symbol by symbol). Can change subtype in middle of code to create shorter barcodes (example 128A vs 128raw). Startcode need to be specified (103-105 for A/B/C). Checksum digit: automatic.	
EAN128 C [A/B/C]	EAN128 (A, B, and C). Specified variant of Code 128, utilising an FNC1 start code. Also known as UCC/EAN-128 or GS1-128. Valid characters: [cf. Code 128]. Checksum digit: automatic.	
C39	CODE 39 - ANSI MH10.8M-1983 - USD-3 - 3 of 9. Valid characters: [0-9 A-Z ' ' . Space \$/+%]	
C39+	CODE 39 + CHECKSUM. Checksum digit: automatic.	
C39E	CODE 39 EXTENDED. Valid characters: [ASCII-characters between 0..127]	
C39E+	CODE 39 EXTENDED + CHECKSUM. Checksum digit: automatic.	
S25	Standard 2 of 5. Valid characters: [0-9]	
S25+	Standard 2 of 5 + CHECKSUM. Valid characters: [0-9]. Checksum digit: automatic.	
I25	Interleaved 2 of 5. Valid characters: [0-9]	
I25+	Interleaved 2 of 5 + CHECKSUM. Valid characters: [0-9]. Checksum digit: automatic.	
I25B	Interleaved 2 of 5 with bearer bars. Valid characters: [0-9]	
I25B+	Interleaved 2 of 5 + CHECKSUM with bearer bars. Valid characters: [0-9]. Checksum digit: automatic.	
C93	CODE 93 - USS-93 (extended). Valid characters: [ASCII-characters between 0..127]. Checksum digits: automatic.	
MSI	MSI. Modified Plessey. Valid characters: [0-9]	
MSI+	MSI + CHECKSUM (module 11). Checksum digit: automatic.	

CODE	DESCRIPTION	BARCODE
CODABAR	CODABAR. Valid characters: [0-9 '- \$:/.+ ABCD] ABCD are used as stop and start characters e.g. A34698735B	
CODE11	CODE 11. Valid characters: [0-9 and '-]. Checksum digits: 1 (or 2 if length of code is > 10 characters) - automatic.	

2D Code (QR-code)

A nominal height and width for these barcodes is set as 25mm. 'size' will scale both the height and width.

CODE	DESCRIPTION	BARCODE
QR-code	QR-code. Accepts numeric, alphanumeric or binary input, with specified error level: L, M, Q, H.	
QR-code with line break	QR-code code can contain \n or \r\n sequences that will be converted to LF or CRLF characters	

Useful links

<http://www.adams1.com>

http://www.tec-it.com/Download/PDF/Barcode_Reference_EN.pdf

<http://www.tec-it.com/en/support/knowledge/symbologies/barcode-overview/linear/Default.aspx>

http://www.gs1uk.org/downloads/bar_code/Bar%20coding%20getting%20it%20right.pdf

<http://web.archive.org/web/19990501035133/http://www.uc-council.org/d36-d.htm> (EAN2 and EAN5)

<http://www.barcodeisland.com/ean13.phtml> (UPC-A)

<http://www.idautomation.com/fonts/postnet/#Specifications>

http://www.outputlinks.com/sites/AFP/ibm_bcocafaq.pdf

https://ribbs.usps.gov/intelligentmail_mailpieces/documents/tech_guides/USPSIMB_Tech_Resource_Guide.pdf (Intelligent Mail)

http://www.mailsorttechnical.com/downloads_mailsort_user_guide.cfm

http://www.mailsorttechnical.com/docs/mug_jun_2009/MUG_10_2008_Mailsort_700.pdf page 20

Human-readable text

Human-readable text is only produced as part of the barcode object in EAN-13, ISBN, ISSN, EAN-8, UPC-A and UPC-E. Here is an example to add text to a barcode:

```
<div style="position:fixed; right: 50mm; top: 60mm; border: 0.2mm solid #000000; text-align: center; padding: 0.5mm; padding-top: 2mm;">  
<barcode code="00034698735346987355" type="EAN128C" /><br />  
<div style="font-family: ocrb;">(00) 0346987 35346987 355</div>  
</div>
```

